

Fundamentals of Information Systems Security

Cryptography Algorithms Comparison

Version: 1.0Quick Reference Guide

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1 Symmetric Encryption Algorithms

1.1 Block Ciphers

Algorithm	Key Size	Block Size	Status	Use Cases
AES	128, 192, 256	128 bits	Secure (Standard)	FIPS, TLS, Disk Encryption
3DES	168 bits	64 bits	Deprecated	Legacy Systems (Avoid)
DES	56 bits	64 bits	Insecure	Historical Only
Blowfish	32-448 bits	64 bits	Legacy	Legacy Systems
Twofish	128, 192, 256	128 bits	Secure	Open Source, PGP
Serpent	128, 192, 256	128 bits	Secure	Disk Encryption
Camellia	128, 192, 256	128 bits	Secure	Japanese Standard, TLS
IDEA	128 bits	64 bits	Legacy	PGP (Older Versions)

1.2 Stream Ciphers

Algorithm	Key Size	Status	Use Cases
RC4	40-2048 bits	Insecure	WEP, SSL/TLS (Avoid)
Salsa20	128, 256 bits	Secure	ChaCha20 Family, TLS
ChaCha20	256 bits	Highly Secure	TLS 1.3, OpenSSH, Disk
AES-CTR	128, 192, 256	Secure	TLS, Cloud Encryption

2 Asymmetric Encryption Algorithms

Algorithm	Key Size	Used For	Status	Security Notes
RSA	2048-4096 bits	Encryption, Signing	Secure	Most common, factoring-based
ECC	256-521 bits	Encryption, Signing	Highly Secure	Smaller keys, faster
ElGamal	2048+ bits	Encryption, Signing	Secure	Diffie-Hellman variant
DSA	2048 bits	Digital Signatures	Secure	NIST-approved
ECDSA	256-521 bits	Digital Signatures	Highly Secure	ECC-based signatures
EdDSA	256 bits	Digital Signatures	Highly Secure	Ed25519, Ed448

3 Hash Functions

Algorithm	Output Size	Status	Use Cases
MD5	128 bits	Broken	Checksums (Never for security)
SHA-1	160 bits	Broken	SSL Certificates (Avoid)
SHA-224	224 bits	Secure	NIST Standard
SHA-256	256 bits	Highly Secure	Most Common, TLS, Bitcoin
SHA-384	384 bits	Highly Secure	NIST Standard
SHA-512	512 bits	Highly Secure	Most Secure SHA-2
SHA-3 (Keccak)	224, 256, 384, 512	Secure	NIST Standard
bcrypt	Variable (up to 448)	Highly Secure	Password Hashing
scrypt	Variable	Highly Secure	Password Hashing, Memory-hard
Argon2	Variable	Highest	Password Hashing (Winner PHC)

4 Cryptographic Protocols

Protocol	Primarily Uses	Status	Application
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SSL 3.0	RSA, DES, 3DES	Deprecated	None (Insecure)
TLS 1.2	AES, RSA, ECDHE	Secure	HTTPS, Email, VPN
TLS 1.3	ChaCha20, EdDSA, AEAD	Highly Secure	Modern HTTPS, QUIC
IPSec	AES, SHA, RSA	Secure	VPN, Network Security
SSH	AES, RSA, ECDSA	Secure	Secure Shell, SFTP
OpenPGP	AES, RSA, DSA	Secure	Email Encryption, Signing
S/MIME	AES, RSA, SHA	Secure	Email Encryption

5 Key Exchange Protocols

Protocol	Used In	Status	Strengths
Diffie-Hellman (DH)	TLS, SSH, IPSec	Secure	First Public Key Exchange
ECDH	TLS 1.3, SSH, ECC	Highly Secure	Smaller Keys, Faster
DHE (Ephemeral DH)	TLS	Secure	Forward Secrecy
ECDHE	TLS 1.3	Highly Secure	Forward Secrecy + ECC

6 Cryptographic Comparison Table

Quick Comparison				
Feature	Symmetric	Asymmetric	Hash	Protocol
Speed	Fastest	Slow	Very Fast	Slow
Key Length	Short (128-256 bits)	Long (2048+ bits)	N/A	Variable
Key Distribution	Hard	Easy	N/A	Easy (via PKI)
Confidentiality	Yes	Yes	No	Yes
Integrity	Yes (with MAC)	Yes (with signing)	Yes	Yes
Authentication	No	Yes (via PKI)	No	Yes
Non-Repudiation	No	Yes (with signing)	No	Yes
Examples	AES, ChaCha20	RSA, ECC	SHA-256, SHA-3	TLS 1.3, SSH

7 Use Case Recommendations

When to Use Each Type

- **Bulk Data Encryption:** Use Symmetric (AES-256-GCM or ChaCha20-Poly1305)
- **Key Exchange:** Use Asymmetric (ECDHE for perfect forward secrecy)
- **Digital Signatures:** Use Asymmetric (EdDSA or ECDSA)
- **Password Storage:** Use Hash (bcrypt, scrypt, or Argon2)
- **Data Integrity:** Use Hash (SHA-256 or SHA-3)
- **Secure Communication:** Use TLS 1.3 with ECDHE and AES-GCM

8 Common Weaknesses

Known Cryptographic Weaknesses

Algorithm	Weakness	Status
MD5	Collision attacks	Broken
SHA-1	Collision attacks (SHAttered)	Broken
DES	56-bit key (brute force)	Insecure
3DES	64-bit block (Sweet32)	Deprecated
RC4	Weaknesses in key schedule	Insecure
RSA (1024-bit)	1024-bit factoring	Insecure
SSL 3.0	POODLE attack	Deprecated
TLS 1.0, 1.1	Multiple vulnerabilities	Deprecated